RAJALAKSHMI ENGINEERING COLLEGE

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A2

Started on	Thursday, 17 September 2026, 10:41 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:50 AM
Time taken	8 mins 58 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int findFloor(int arr[],int low,int high,int x){
    if(low>high)
    return-1;
    int mid=low+(high-low)/2;
    if(arr[mid]==x){
        return arr[mid];
    }
    if(arr[mid]>x){
        return findFloor(arr,low,mid-1,x);
    }
    int result=findFloor(arr,mid+1,high,x);
    if(result==-1)
    return arr[mid];
    return result;
}
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d", &arr[i]);
    }
    int x;
    scanf("%d",&x);
    int floor=findFloor(arr,0,n-1,x);
    printf("%d\n", floor);
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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